Gaurav Kumar Choudhary

Ravangla, Namchi, Sikkim (available for a 6-month in-office internship in Noida) | +91-8340459246 | gauravgkchy139@gmail.com |
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PROFILE

Final-year B.Tech Computer Science student at NIT Sikkim (CGPA 9.17/10) who builds AI-first web products end to end: **React** front ends, **Python (FastAPI, Flask)** and **Java (Spring Boot)** REST back ends on **PostgreSQL** and **MySQL**, and LLM features on top of them (**RAG assistants, tool-calling AI agents, MCP servers**). Two internships on production software: fixed reliability defects and shipped a full feature on a live LLM voice-agent platform, and lifted an LLM document classifier from 70% to 85% accuracy. Daily user of AI coding assistants (Claude Code, ChatGPT). LeetCode Knight (top 5%, 500+ solved); GATE 2026 CS AIR 8211.

EDUCATION

National Institute of Technology, Sikkim — Aug 2023 - May 2027

B.Tech in Computer Science and Engineering — *CGPA: 9.17/10.0*

Class XII (CBSE, PCM): 86.6% | **Class X (CBSE)**: 90.6% — 2023 | 2021

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, SQL, C++, Bash
Frontend: React.js, Next.js, HTML, CSS, responsive layouts, dashboards over REST and streamed (SSE) data
Backend and APIs: FastAPI, Flask, Spring Boot, REST API design, JWT authentication and role-based access control, WebSockets, Server-Sent Events, async I/O
Databases: PostgreSQL (migrations, triggers, row-level security, connection pooling, query tuning), MySQL, Redis, MongoDB
Generative AI: LLM APIs (OpenAI-compatible clients, Llama-3 via Groq, Qwen3), prompt engineering, function calling and structured JSON output; RAG (chunking, embeddings, SentenceTransformers, FAISS); AI agents and tool-calling loops, Model Context Protocol (MCP) servers, self-hosted OpenClaw; PyTorch, Hugging Face Transformers
Tools: Git, GitHub (pull requests and code review), Docker, Docker Compose, GitHub Actions CI/CD, Linux, VS Code, Postman, AWS S3

EXPERIENCE

Software Engineering Intern (Backend and QA) — Jun 2026 - Aug 2026

Blessing Softech Pvt. Ltd.; Fonazo AI Voice Platform (production LLM voice agent) — *Remote*

- Shipped **agent configuration versioning and rollback** end to end: a schema migration, a database trigger that snapshots every config change into append-only history, row-level security, and **GET /versions** and **POST /rollback** REST APIs that re-validate before restoring; kept the 181-test regression suite green through a full code-review cycle.
- Root-caused and fixed three production defects on the live **FastAPI and PostgreSQL** service: asyncpg connection-pool exhaustion that dropped calls under load (pool 5 to 20, 30s timeout), a Redis lookup blocking about 2s when the cache was down (made fail-open), and an unhandled WebSocket exception that silently killed the request path.
- Measured the end-to-end **latency budget** of the AI pipeline (speech-to-text about 800ms, LLM time-to-first-token about 1.2s, speech synthesis about 600ms, warm RAG retrieval about 190ms) and isolated embedding-model cold start as the dominant first-call cost.
- Owned the release gate in weekly sprints: regression runs, severity-rated defect reports, **prompt-injection and PII-disclosure testing** of the LLM agent; drove a FAISS retrieval bug in the RAG path to resolution before production sign-off.

Software Development Engineering Intern — Dec 2025 - Mar 2026

BLG Technologies Pvt. Ltd. — *Remote*

- Integrated **LLM-based classification** for tender eligibility screening, lifting accuracy from **70% to 85%** through iterative prompt and pipeline tuning, with a human-in-the-loop path that routes low-confidence predictions to reviewers.
- Built **document intelligence** tooling that parsed and normalized **500+ unstructured documents** into standardized binders (reviewer time down 60%), served through **Flask REST APIs on PostgreSQL** with React components for filtering, automating the submission, review and sign-off workflow and cutting manual effort by 40%.

PROJECTS

Context-Aware AI Assistant (RAG) | *Python, FastAPI, SentenceTransformers, FAISS, Llama-3 (Groq), Docker* — GitHub

- Built and containerized an AI assistant behind a browser extension: chunking, embeddings and FAISS vector search ground every answer in the page the user is reading (retrieval under 3s), with the LLM response streamed token by token over Server-Sent Events from an async FastAPI back end.

AI Agents: Tool-Calling Loop and MCP Servers | *Python, MCP, OpenClaw* — Ongoing

- Hand-built a tool-calling agent loop (model call, tool dispatch, result feedback, termination, failure handling when a tool raises or returns junk) with per-step logs so every run can be replayed; wrote **MCP servers** exposing custom tools to agents and self-host OpenClaw with tools and channels wired up.

Library Management System (full stack) | *Java 21, Spring Boot, Spring Security, JPA, MySQL, JWT, React.js* — GitHub

- Modelled an issue-return lifecycle (ISSUED, RETURNED, OVERDUE) across 4 JPA entities with guard conditions rejecting invalid borrows; exposed **30+ REST endpoints** behind stateless JWT authentication and role-based access control, consumed by a **React.js** admin dashboard.

AI Code Review Environment (agent evaluation) | *Python, Pydantic, OpenAI-compatible client, Docker* — GitHub

- Built an OpenEnv environment that scores LLM code-review agents: typed reset/step/state API, three deterministic graders (bug fix, optimization, security refactor) with shaped rewards, a baseline inference script with structured step logs, and a Docker image that passes validation.

Steward: Memory-Aware LLM Inference (first-author research) | *Python, PyTorch, Hugging Face, Qwen3-8B* — In progress

- Built the evaluation pipeline and a controller that allocates GPU memory across concurrent LLM requests: on 480 real requests, up to +15.4 F1 over uniform allocation with zero out-of-memory failures; caught that 4-point effects at n=50 vanished at n≥150 and rebuilt every conclusion on n≥100 per cell.

ACHIEVEMENTS AND LEADERSHIP

- **Competitive programming:** LeetCode Knight, top 5%, 500+ problems solved; GATE 2026 Computer Science All India Rank 8211 among 200,000+ candidates.
- **Certifications:** NPTEL Elite, Introduction to Large Language Models, IIT Madras (top 5%); McKinsey Forward Programme completed.
- **Leadership:** Devcans Coding Club, NIT Sikkim: organized workshops and hackathons for 100+ students; mentored peers in DSA and backend development.